



Case-Control Studies

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Retrospective Cohort vs. Case-Control

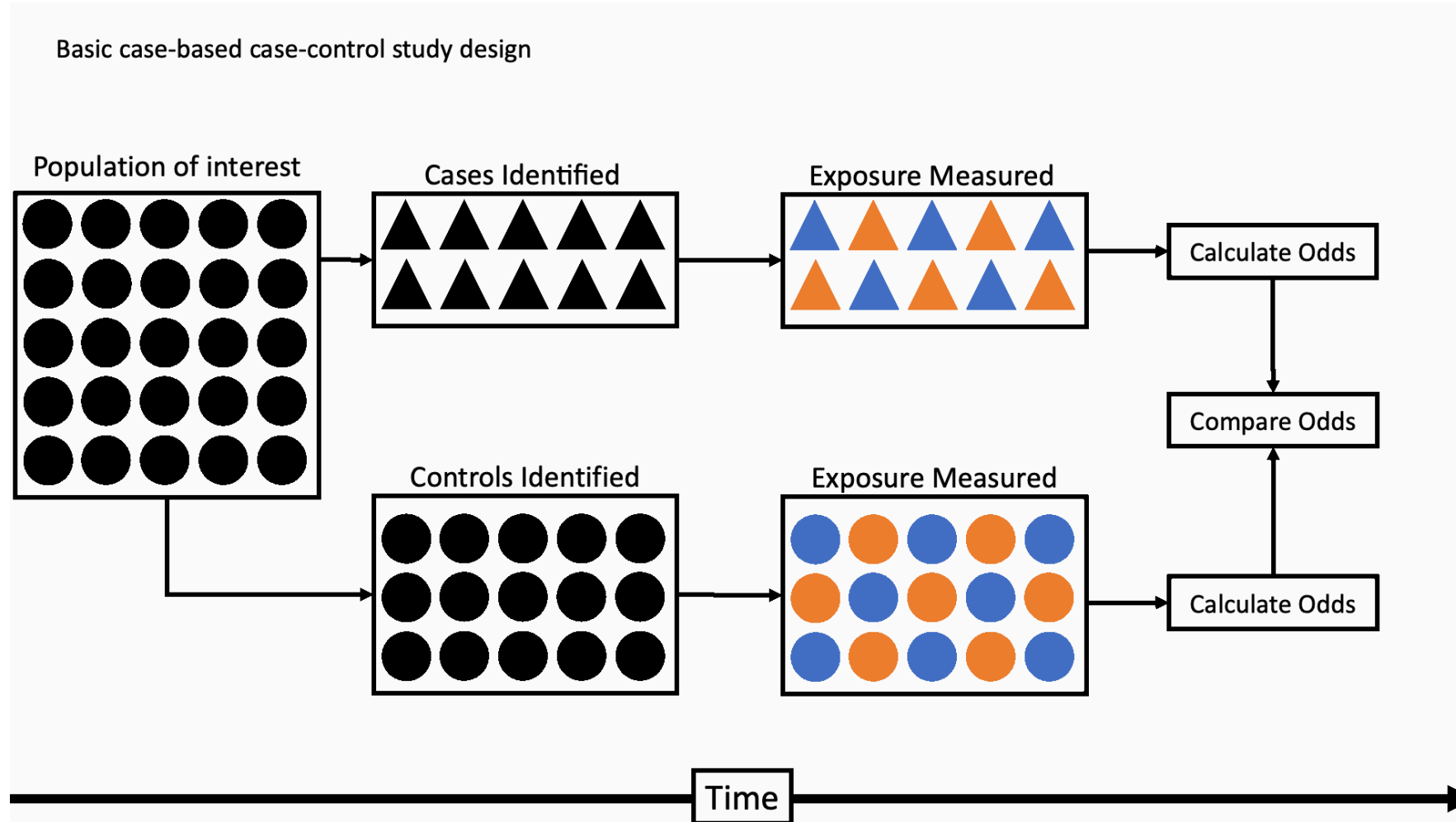
- Case-control
 - Sample based on outcome and look backward for exposure.
- Retrospective Cohort
 - Sample based on exposure and look “forward” towards the outcome
 - A cohort study was done, you just weren’t there to initially observe it

When Cohort Is Not Ideal

Cohort studies are powerful — but not always practical.

- What if:
 - The disease is **rare**?
 - The outcome has a **long latency period**?
 - Long-term follow-up is **too costly or infeasible**?
 - You need results **quickly**?
 - In these situations, a different design may be more efficient.

Basic Case-Control Study Design



Case-Control Studies: Overview

- **Definition:** Start with **outcome status**
 - **Cases** → Have the disease; **Controls** → Do not have the disease
 - Then look backward to assess **prior exposure**
- **Direction of Inquiry:** Outcome → Exposure
- **Primary Measure of Association: Odds Ratio (OR)**

Poll: Identify the Study Design

- Researchers identify 500 patients with lung cancer and 500 patients without lung cancer. They compare prior smoking history.
 - What type of study is this?
- A.** Cohort study
 - B.** Case-control study
 - C.** Cross-sectional study
 - D.** Randomized trial

Discussion: How Would You Study Lung Cancer?

If lung cancer is a **rare disease**, how would you design the study?

- Would you start with **exposure** or **outcome**?
- Would long-term follow-up be practical?
- How could you efficiently identify cases?
- What comparison group would you need?

Doll and Hill (1950)

- **Research Question:** Is cigarette smoking associated with lung cancer?
- **Study Design:**
 - **Cases:** Patients diagnosed with lung cancer
 - **Controls:** Hospital patients without lung cancer
 - Collected prior **smoking history**
- **Direction of Inquiry:** Outcome → Exposure
- **Why Case-Control Was Appropriate:**
 - Lung cancer was a **rare disease**
 - Long-term follow-up would have been impractical
 - Efficient comparison of prior exposure

Review: Population and Sample

Example: We want to understand the health of adults living in Texas.

Group	Who belongs to it?	In our example
Target population	People we want findings to apply to	All Texas adults
Source population	People from whom we select participants	Adult patients at several Texas clinics
Study sample	People whose data we analyze	500 participating patients

- **Chance:** Would another random sample from the same source give a different estimate?
- **Applicability:** How well do these findings apply to all Texas adults?

Study Base Principle

- Cases and controls should be “representative of the same base experience.”
- If a potential control had gotten the disease, they would have been selected with the same probability as any other case
 - If this is not true, you have selection bias!

Group Exercise: Shands Hospital

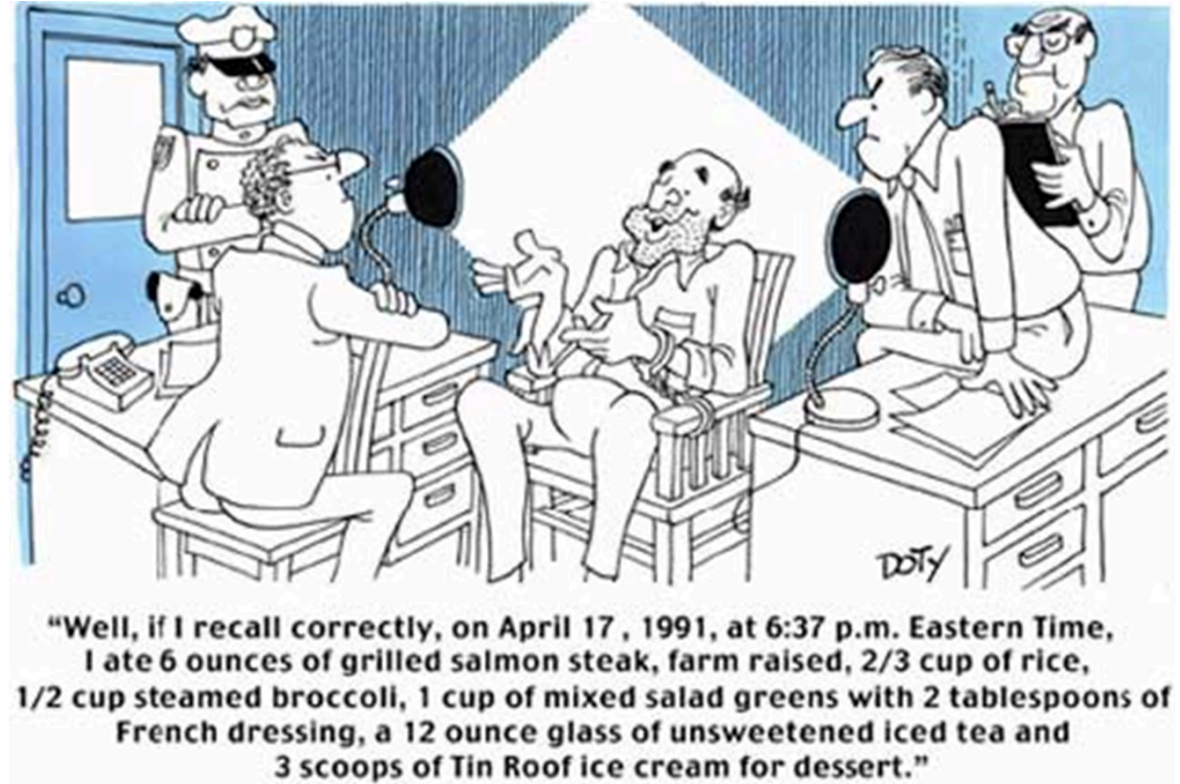
Shands Hospital at the University of Florida is a referral hospital for a variety of cancer diagnoses -- including bronchogenic carcinoma -- for the North Florida region where it is located. Researchers are interested in the association between bronchogenic carcinoma and a history of cigarette smoking, and they decide to conduct a case-control study. Cases are selected from the admissions records to the cancer center. Controls are randomly selected from the population of the surrounding county. Does the control group violate the study base principle?

Selection Bias in Case-Control Studies

- **Core Issue:** Controls must represent the **same source population** as the cases.
 - If not, the exposure comparison becomes biased.
- **Example (Smoking & Lung Cancer):**
 - If controls are selected from a group with unusually low smoking rates, smoking may appear more strongly associated with lung cancer.
- **Key Principle:**
 - Controls should reflect the **exposure distribution** of the population that produced the cases.

Recall Bias

- **Definition:** A type of information bias in which participants remember past exposures **differently**.
- **When it occurs**
 - Often in **case-control studies**
 - Cases may recall exposures more carefully than controls
- **Key idea:** Differences in memory between groups can distort the exposure–outcome association.



Interviewer and Observer Bias

- **Interviewer Bias**

- Occurs during **data collection**
- Interviewer knowledge influences how questions are asked
- Can lead to different responses between groups

- **Observer Bias**

- Occurs during **outcome assessment**
- Observer expectations influence how outcomes are recorded
- More likely when blinding is absent

- **Key idea:** When researchers influence data collection or measurement, results may be systematically distorted.

Why Use an Odds Ratio?

- In case-control studies, we cannot calculate **risk** directly. Why?
 - We start with a fixed number of **cases and controls**
 - The total population at risk is unknown
 - Incidence cannot be measured
 - Instead, we quantify association using the **Odds Ratio (OR)**
- **What Does OR Measure?**
 - The **odds of prior exposure** among cases compared to the odds of exposure among controls
- **Key Contrast:**
 - Cohort → Uses **Relative Risk (RR)**
 - Case-Control → Uses **Odds Ratio (OR)**

Calculating Odds Ratio

	Cases	Controls
Exposed	a	b
Not exposed	c	d
Odds of exposure	a/c	b/d

Odds ratio: odds of exposure in cases versus odds of exposure in controls.

$$\text{OR} = (a/c) \div (b/d) = (a \times d)/(b \times c)$$

Smoking and Lung Cancer

- **Research Question:** Is smoking associated with lung cancer?
- **Study Design:**
 - **Cases:** Patients with lung cancer
 - **Controls:** Patients without lung cancer
 - Collected prior **smoking history**
- **OR** = $(A \times D) / (B \times C) = (647 \times 27) / (622 \times 2) \approx 14.0$
- **Interpretation:** The odds of prior smoking among lung cancer cases are about **14 times higher** than among controls.
- **Conclusion:** Smoking is strongly associated with lung cancer.

	Lung Cancer (+)	Lung Cancer (–)
Smoker	647	622
Non-Smoker	2	27

Interpreting the Odds Ratio

- **Interpreting the Odds Ratio (OR)**
 - **OR = 1:** No association between exposure and disease
 - **OR > 1:** Exposure is associated with **higher odds** of disease
 - **OR < 1:** Exposure is associated with **lower odds** of disease (protective effect)
- **Important:** OR reflects **odds**, not risk.
- When the disease is **rare**, the OR closely approximates the **Relative Risk (RR)**.
 - When cases are rare: **Total population $\approx$ Non-cases**

Poll: Interpreting Odds Ratio

- In a case-control study examining asbestos exposure and lung cancer, researchers identified 400 lung cancer cases and 400 controls without lung cancer. The Odds Ratio (OR) for asbestos exposure was **3.0**.
- What does this mean?
 - A.** Exposed individuals have 3 times the risk of lung cancer
 - B.** Asbestos exposure causes lung cancer
 - C.** There is no association between asbestos and lung cancer
 - D.** Lung cancer cases had 3 times the odds of prior asbestos exposure compared to controls

Poll: Identifying Bias

- In a case-control study investigating the relationship between alcohol use and liver cancer, cases were recruited from a major urban hospital. Controls were recruited from a local fitness club.
 - Which bias is most likely?
- A.** Recall bias
 - B.** Attrition bias
 - C.** Selection bias
 - D.** Interviewer bias

Strengths and Limitations

Strengths

- Efficient for studying **rare diseases**
- Faster and less costly than cohort studies
- Require **fewer participants**
- Useful for diseases with **long latency**

Limitations

- Cannot directly measure **incidence or risk**
- Vulnerable to **selection bias**
- Prone to **recall bias**
- Control selection can be challenging

Cohort vs. Case-Control Studies

Feature	Cohort Study	Case-Control Study
Start With	Exposure	Outcome
Direction	Exposure → Outcome	Outcome → Exposure
Measure	Relative Risk (RR)	Odds Ratio (OR)
Best For	Rare exposures	Rare diseases
Time & Cost	Longer, more expensive	Faster, less expensive
Main Bias Concern	Loss to follow-up	Selection & recall bias
Can Measure Incidence?	Yes	No